

A VERSATILE FREQUENCY COUNTER

Often, the need for a frequency counter is felt that operates in the LF, HF and VHF ranges. The most common deterrents are the lack of programming knowledge and the non-availability of VHF prescalers to bring down VHF to a manageable HF range. Here we show how a commonly available low-cost radio may be ripped apart and used as a frequency counter covering frequencies from 500 kHz to 150 MHz

■ VASUKI PRASAD

Chinese radio company Kchibo makes radios based on the CX-A1619BM IC. This IC is a surface-mount version of the CXA1619BS, which most designers are familiar with; FM boards available in the market are built using this IC. Kchibo and other companies went a step ahead and thought of implementing a radio frequency display, along with an alarm clock, in their radios so that any station can be tuned into, without doubting the actual frequency which the radio is tuned to (no blind or approximate tuning). This is not a radio synthesiser; it's a conventional, mechanically tuned radio that sports a frequency display.

As opposed to conventional radios which use a single IC (the CXA1619BS/BM), the so-called digital radio uses two ICs: the radio receiver IC and the UTC6610.

At the heart of this project is UTC6610. It comes in many packages, but these radios use the chip-on-board variety. The UTC6610 comes equipped with a frequency counter circuitry, prescaler for VHF, clock circuitry (time), on-chip oscillator, simple adder/subtractor or offset unit, and LCD driver. As can be seen, this unit is completely self-sufficient. The only external components required are crystal and LCD.

The frequency counter circuitry of the IC measures the frequency of the incoming RF wave by counting the number of transitions (high-to-low or

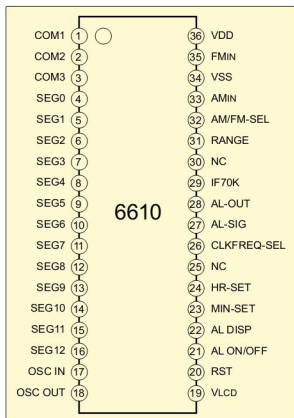


Fig. 1: Pin configuration of UTC6610

low-to-high) within a set period.

The prescaler for VHF is a divide-by-ten block that probably uses high-speed CMOS circuitry to divide the number of pulses. When a 150MHz signal is fed to the prescaler, the output is 15 MHz. Datasheets specify 300mV peak-to-peak input; higher voltages will destroy the IC. The prescaler is activated only in the FM mode of operation.

As regards clock circuitry, we're not interested in it and hence skipping

this section. It is used only to keep track of time.

The on-chip oscillator is a NOT gate that requires an external 32.768kHz crystal to generate the timing base.

The adder/subtractor unit works like this: A super heterodyne radio receiver requires a local oscillator operating at a fixed frequency above the incoming signal's frequency. This fixed frequency is called intermediate frequency, or IF. For commercial FM radios, it is 10.7 MHz for FM and 455 kHz for AM/SW.

Now, if the frequency counter was reading the actual frequency of the oscillator, the user would have been put to

great trouble in determining the actual receiving frequency. For example, for a 100MHz signal to be detected, the local oscillator is made to run at 110.7 MHz ($100+10.7=110.7$ MHz). So to display the intended receiving frequency, the UTC6610 subtracts 10.7 MHz from the counted frequency. This means that if you are to use the UTC6610 as a frequency counter, always bear in mind that the actual frequency is 10.7 MHz higher than the displayed frequency